



Data Visualization Associate Early Internship

SLU 0408 DVA Team 40

3rd Week Deliverables Report

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Week 3 – Development of Data Visualization in Looker Studio (Team Report)

Internship: Data Visualization Associate (Excelerate Virtual Internship)

Week: 3 – Understanding Master_Table, Mapping Table, Wireframe Design & Dashboard Planning.

Team Contribution: Joint report covering mapping table creation, KPI identification, and wireframe development for dashboard visualization.

Introduction

In Week 2 of the Excelerate Data Visualization Internship, the team focused on consolidating multiple raw datasets into a single structured Master Table through an automated ETL process. This involved designing the schema, defining constraints and keys, developing a stored procedure to extract, transform, and load data, and applying cleaning rules to handle duplicates, missing values, and inconsistent text formats. By the end of Week 2, we successfully delivered a high-quality Master Table, an automated ETL procedure, and a Data Quality Report, establishing a solid foundation for subsequent visualization and analytical tasks.

Building on this work, in Week 3 we will focus on preparing the groundwork for developing an impactful dashboard in Looker Studio. We will create a comprehensive Mapping Table to align key data points across different sources, define column mappings, data transformations, and apply business logic to ensure visualization readiness. We will identify key metrics and KPIs to enable effective data storytelling. Additionally, we will design a detailed wireframe—either digital or hand-drawn—to structure the dashboard layout, including placement of charts, tables, KPIs, and interactive filters. Annotations will be added to the wireframe to explain the purpose of each section and the reasoning behind visualization choices. These efforts will ensure that all data is organized, metrics are clearly defined, and the dashboard design is structured for clarity, usability, and impactful insights in the upcoming development stage.

Objectives

- Create a Mapping Table aligning key data points across different sources, including column mappings, data transformations, and applied business logic for visualization readiness
- Identify key metrics and KPIs for impactful data storytelling on the dashboard
- Design a wireframe (digital or hand-drawn) outlining the dashboard structure, including placement of KPIs, charts, tables, and filters
- Add annotations to the wireframe explaining the purpose of each section and the reasoning behind layout and visualization choices

Datasets Review :

Category	Datasets
Assigned Datasets	1. User_data.csv 2. Opportunity_Raw.csv 3. 3. Cohort_Raw.csv 4. 4. Marketing_data.csv 5. 5. Learner_Opportunity_Raw.csv 6. 6. Cognito_Raw2.csv
Chosen Datasets for Master Table	1. User_data.csv 2. Opportunity_Raw.csv 3. Cohort_Raw.csv 4. Learner_Opportunity_Raw.csv 5. Cognito_Raw2.csv

Expected Deliverables by the End of the Week

- ❖ A comprehensive Mapping Table ensuring data consistency and readiness for visualization
- ❖ A clear and structured wireframe with annotations, serving as a blueprint for the dashboard
- ❖ A well-planned layout of visual elements with clarity and usability, setting the stage for an interactive and insightful dashboard experience

1. Understanding Master_Table

Column Name	Datatype	Description
learner_id	TEXT	Unique identifier for each learner. Ensures consistent linking of learner data across systems.
opportunity_id	TEXT	Unique identifier for each opportunity. Useful for tracking and differentiating multiple programs.
assigned_cohort	TEXT	The cohort assigned to the learner. Helps group learners into batches for analysis.
apply_date	Timestamp with time zone	The date and time when the learner applied. Enables time-based reporting on applications.
status	TEXT	The application status of the learner (e.g., Enrolled, Applied). Useful for monitoring progress and outcomes.
country	TEXT	The country of the learner. Supports global reach and regional distribution analysis.
degree	TEXT	The degree the learner is pursuing or has obtained. Useful for academic segmentation.
institution	TEXT	The educational institution the learner is associated with. Helps identify institutional participation trends.
major	TEXT	The academic major of the learner. Enables program alignment with field-specific learning needs.
email	Character Varying	The email address of the learner. Acts as a key contact point for communication and recovery.
gender	TEXT	The gender of the learner. Assists in demographic breakdowns and diversity tracking.
user_create_date	Timestamp with time zone	The date and time when the user profile was created. Helps measure the start of learner engagement.
user_last_modified_date	Timestamp with time zone	The date and time of the last modification to the user profile. Useful for tracking updates and recent activity.
birthdate	Date	The date of birth of the learner. Supports age-based eligibility and segmentation.
city	TEXT	The city of residence of the learner. Enables geographic insights at a local level.
state	TEXT	The state or province of residence of the learner. Adds more regional detail for analysis.
opportunity_name	Character Varying	The name of the opportunity (e.g., Course, Internship). Provides descriptive context of the program.
opportunity_category	Character Varying	The category of the opportunity (e.g., Course, Internship). Helps classify offerings by type.
opportunity_code	Character Varying	A unique code for the opportunity. Simplifies backend

Column Name	Datatype	Description
		tracking and reporting.
cohort_start_date	Timestamp with time zone	The start date of the cohort. Marks the beginning of the learning period.
cohort_end_date	Timestamp with time zone	The end date of the cohort. Defines when the opportunity concludes.
cohort_size	Integer	The number of learners in the cohort. Useful for resource allocation and capacity planning.

2. Identification of Important Columns in Master_Table

KPI Specification for Master Table

Derived Fields (for KPIs)

- **application_date:** DATE(apply_date)
- **account_created_date:** DATE(user_create_date)
- **age_at_apply:** DATE_PART('year', AGE(apply_date, TO_DATE(birthdate,'DD-MM-YYYY')))
- **cohort_duration_days:** cohort_end_date::date - cohort_start_date::date
- **time_to_start_days:** cohort_start_date::date - apply_date::date
- **account_to_apply_days:** apply_date::date - user_create_date::date
- **email_domain:** LOWER(SPLIT_PART(email,'@',2))

KPI List (with pseudo formulas)

1) Total Applications

- Definition: Count of application records created in the selected period.
- Formula: COUNT(*) (filtered by apply_date in period)
- Indicates: Overall demand across opportunities.
- Dashboard use: High-level demand trend; compare periods.

2) Unique Learners

- Definition: Distinct people who applied.
- Formula: COUNT(DISTINCT learner_id)
- Indicates: Reach; mitigates duplicate counts from multi-application learners.
- Dashboard use: Marketing reach vs. application volume.

3) Applications per Opportunity

- Definition: Load per program/cohort.
- Formula: COUNT(*) BY opportunity_id
- Indicates: Program popularity & funnel top.
- Dashboard use: Rank opportunities by demand.

4) Enrollment Count

- Definition: Number of records with status = 'Enrolled'.
- Formula: SUM(CASE WHEN status='Enrolled' THEN 1 ELSE 0 END)
- Indicates: Successful conversions into seats.
- Dashboard use: Core success metric.

5) Enrollment Rate (Application→Enrollment)

- Definition: Share of applications that become enrolled.
- Formula: Enrollment Count / Total Applications
- Indicates: Funnel effectiveness and screening.
- Dashboard use: Track improvements from ops changes.

6) Cohort Capacity (Planned Seats)

- Definition: Sum of planned seats for cohorts within scope.
- Formula: SUM(DISTINCT cohort_size BY opportunity_id) or SUM(cohort_size) in a cohort-level table
- Indicates: Supply (seat availability).
- Dashboard use: Capacity planning.

7) Cohort Fill (Enrolled Seats)

- Definition: Enrolled headcount per cohort.
- Formula: COUNT(*) WHERE status='Enrolled' GROUP BY opportunity_id
- Indicates: Actual seats taken.
- Dashboard use: Monitor fill progress.

8) Capacity Utilization (%)

- Definition: Filled seats vs. planned seats.
 - Formula: Cohort Fill / cohort_size
 - Indicates: Over/under-subscription risk.
 - Dashboard use: Highlight cohorts needing promotion or caps.
- 9) Seats Remaining
- Definition: Vacant seats per cohort.
 - Formula: GREATEST(cohort_size - Cohort Fill, 0)
 - Indicates: Sell-through potential.
 - Dashboard use: Prioritize marketing for low-fill cohorts.
- 10) Application Velocity (Daily Avg)
- Definition: Average applications per day in period.
 - Formula: COUNT(*) / DATE_DIFF('day', period_start, period_end)
 - Indicates: Demand momentum.
 - Dashboard use: Forecast pipeline health.
- 11) Time to Cohort Start (Days)
- Definition: Days between application and cohort start.
 - Formula: AVG(time_to_start_days)
 - Indicates: Lead time learners give; informs campaign timing.
 - Dashboard use: Optimize outreach windows.
- 12) Account→Application Lag (Days)
- Definition: Days from account creation to first application.
 - Formula: AVG(account_to_apply_days); first application per learner
 - Indicates: UX friction or nurture effectiveness.
 - Dashboard use: Improve onboarding and CTAs.
- 13) Active Cohorts
- Definition: Cohorts currently running in the selected date.
 - Formula: COUNT(DISTINCT opportunity_id) WHERE current_date BETWEEN cohort_start_date AND cohort_end_date
 - Indicates: Operational load.
 - Dashboard use: Staffing & support planning.
- 14) Average Cohort Duration (Days)
- Definition: Mean duration of cohorts.
 - Formula: AVG(cohort_duration_days)
 - Indicates: Program design length trends.
 - Dashboard use: Benchmark by category.
- 15) Category Mix (Applications/Enrollments by opportunity_category)
- Definition: Distribution across Course / Internship / Competition, etc.
 - Formula: COUNT(*) BY opportunity_category and Enrollment Count BY opportunity_category
 - Indicates: Portfolio demand & success by type.
 - Dashboard use: Investment allocation.
- 16) Geography Mix & Yield
- Definition: Applications/Enrollments and Enrollment Rate by country/state/city.
 - Formula: COUNT(*), SUM(enrolled), SUM(enrolled)/COUNT(*) BY country[, state]
 - Indicates: High-performing regions and gaps.
 - Dashboard use: Geo-targeted campaigns.
- 17) Institution Mix & Yield
- Definition: Top institutions by applications, enrollments, and conversion.
 - Formula: COUNT(*), SUM(enrolled), SUM(enrolled)/COUNT(*) BY institution
 - Indicates: Campus partnerships performance.
 - Dashboard use: Focus outreach and partnerships.
- 18) Degree/Major Mix
- Definition: Breakdown by degree and major.
 - Formula: COUNT(*) BY degree, major
 - Indicates: Audience profile; program-market fit.
 - Dashboard use: Tailor curriculum & marketing.
- 19) Age Bands at Apply (e.g., <18, 18–22, 23–30, 31+)
- Definition: Applicants grouped by age.
 - Formula: CASE WHEN age_at_apply<18 THEN ... END then COUNT(*) BY band
 - Indicates: Demographic reach and compliance.
 - Dashboard use: Persona targeting; check minors handling.
- 20) Gender Distribution & Completeness
- Definition: Share of each gender and % Unknown.

- Formula: $\text{COUNT(*) BY gender; unknown_rate} = \text{COUNT(*) WHERE gender IS NULL OR gender='Unknown'} / \text{COUNT(*)}$
 - Indicates: Representation and data quality.
 - Dashboard use: Diversity reporting and form optimization.
- 21) Email Domain Mix
- Definition: Applications by email domain (edu, gmail, etc.).
 - Formula: $\text{COUNT(*) BY email_domain}$
 - Indicates: Student vs. non-student split; comms deliverability.
 - Dashboard use: Channel strategy.
- 22) Repeat Applicant Rate
- Definition: Share of learners applying to >1 opportunity.
 - Formula: $\text{COUNT(DISTINCT learner_id HAVING COUNT(DISTINCT opportunity_id)>1)} / \text{COUNT(DISTINCT learner_id)}$
 - Indicates: Cross-interest; possible friction if high with low conversion.
 - Dashboard use: Personalization & recommendations.
- 23) First-Time vs. Returning Applications
- Definition: Split of applications by learner's application sequence.
 - Formula: For each learner_id, rank by apply_date; first_application = rank=1.
 - Indicates: Growth in new audience vs. re-engagement.
 - Dashboard use: Measure acquisition vs. retention levers.
- 24) Status Funnel Breakdown
- Definition: Count of applications by status across funnel stages.
 - Formula: $\text{COUNT(*) BY status (e.g., Applied, Enrolled, Waitlisted, Rejected, Withdrawn, Unknown)}$
 - Indicates: Bottlenecks and leakage.
 - Dashboard use: Ops follow-up and SLA adherence.
- 25) SLA: Decision Latency (if decision timestamps exist)
- Definition: Time from application to final status decision.
 - Formula: $\text{AVG(decision_date - apply_date)}$ (requires decision_date; else omit)
 - Indicates: Ops speed; applicant experience.
 - Dashboard use: Staffing & process adjustments.
- 26) On-Time Start Coverage (Forecast)
- Definition: % cohorts expected to reach a target fill by T days before start.
 - Formula: $\text{COUNT(cohorts WHERE Cohort Fill at -T days} \geq \text{target\%)} / \text{COUNT(cohorts starting in horizon)}$
 - Indicates: Risk flagging for underfilled cohorts.
 - Dashboard use: Early interventions.
- 27) Application Seasonality Index
- Definition: Relative application volume vs. average.
 - Formula: $\text{(Applications in month)} / \text{(12-month average)}$
 - Indicates: Seasonal peaks.
 - Dashboard use: Budget & capacity planning.
- 28) Data Quality: Missing Critical Fields
- Definition: % records missing any of: country, degree, gender, birthdate.
 - Formula: $\text{COUNT(*) WHERE any_critical_null} / \text{COUNT(*)}$
 - Indicates: Reporting reliability.
 - Dashboard use: Form validation and ETL fixes.
- 29) Data Quality: Invalid Birthdate Format
- Definition: % records with non-parsable birthdate.
 - Formula: $\text{COUNT(*) WHERE NOT birthdate} \sim '[0-3]\text{d-[01]}\text{d-d}\{4\}\$' / \text{COUNT(*)}$
 - Indicates: Risk in age-related reporting.
 - Dashboard use: Data cleaning backlog.
- 30) Compliance: Under-18 Applicants
- Definition: Count and share of applicants under 18 at apply.
 - Formula: $\text{COUNT(*) WHERE age_at_apply} < 18; \text{rate} = / \text{Total Applications}$
 - Indicates: Guardian/consent workflows required.
 - Dashboard use: Compliance alerts and capacity.

Suggested Dashboard Pages & Key Slices

1. Executive Overview — KPIs #1–#9, #13, #15; filters by period, category, country.
2. Funnel & Velocity — KPIs #4, #5, #10, #11, #12, #24, #27.
3. Cohort Capacity Monitor — KPIs #6–#9, #13, #14, #26 with cohort-level table and alerts.
4. Audience & Reach — KPIs #2, #16–#23, #30 with demographic and institution maps/bars.
5. Data Quality & Compliance — KPIs #28–#30 with drill-down to offending records.

Master Table Columns (KPI, Categories & Metrics Review):

KPI (Key Performance Indicators)	Categories (Dimensions)	Metrics
Number of applications	learner_id	duration
Total unique students	email	size
Most Popular Course	institution	apply_date
Batch Occupancy	degree	start_date
Avg batch duration	major	end_date
Avg batch size	opp_id	birthdate
Top city by applicants	opp_name	user_create_date
Application Rate over time	opp_code	—
Major age group	category	—
Avg Monthly Applications	assigned_cohort	—
	gender	—
	city	—
	country	—
	state	—
	status	—

3. Understanding Column Relationships in the Master Table

This section outlines the relationships between the various columns in the **Cleaned_Master_Table.csv** file. This understanding is crucial for building a meaningful and interactive dashboard. The relationships are explained with the help of the business logic defined in the **KPI doc.docx**.

1. Learner Profile

The columns in this category describe the individual learners.

- **learner_id**: This is the unique identifier for each learner. Every row in the table is associated with a **learner_id**, but a single learner can appear in multiple rows if they have applied for more than one opportunity.
- **Demographic Information**: Columns like **country**, **degree**, **institution**, **major**, **gender**, **birthdate**, **city**, and **state** provide demographic information about the learner. These are all directly linked to the **learner_id**. For example, a single **learner_id** will always have the same **birthdate** and **gender** across all their applications.
- **Contact Information**: The **email** column is the primary contact information for a learner and is unique to each **learner_id**. The **email_domain** can be extracted from the email to analyze the distribution of learners from academic vs. commercial email providers.
- **Account Information**: **user_create_date** and **user_last_modified_date** track when a learner created their account and when it was last updated. These are linked to the **learner_id** and are useful for calculating metrics like the time taken from account creation to application (**account_to_apply_days**).

2. Opportunity & Cohort Details

This group of columns provides information about the various opportunities and cohorts.

- **opportunity_id**: This is a unique identifier for each opportunity (e.g., a course, internship, or competition).
- **Opportunity Information**: **opportunity_name** and **opportunity_category** are directly associated with the **opportunity_id**, providing a descriptive name and category for each opportunity.
- **assigned_cohort**: This column links a learner's application to a specific cohort within an opportunity. An opportunity can have multiple cohorts.
- **Cohort Timeline**: **cohort_start_date** and **cohort_end_date** define the duration of each cohort. These dates are used to calculate the **cohort_duration_days**.
- **Cohort Size**: The **cohort_size** column indicates the number of available seats in a cohort, which is essential for monitoring capacity and enrollment rates.

3. Application & Status

These columns track the application process and the status of each application.

- **apply_date**: This is the date when a learner submitted their application for an opportunity. It is a key timestamp for analyzing application trends over time.
- **status**: This column indicates the current status of an application, with values such as 'Enrolled', 'Completed', 'Applied', or 'Unknown'. This is a critical field for calculating core success metrics like **Enrollment Count** and **Enrollment Rate**.
- **Timeliness Metrics**: The relationship between date columns allows for the calculation of important KPIs:
 - **time_to_start_days**: This is the difference between the **cohort_start_date** and the **apply_date**, indicating how far in advance learners are applying for opportunities.
 - **account_to_apply_days**: This is the difference between the **apply_date** and the **user_create_date**, showing how long it takes for a new user to apply for an opportunity.

4. Derived Relationships & KPIs

The relationships between the columns in the Master Table are fundamental for deriving the Key Performance Indicators (KPIs) mentioned in the

KPI doc.docx. Here are a few examples:

- **Total Applications vs. Unique Learners**: By counting all rows, we get the **Total Applications**. By counting the distinct **learner_id**'s, we get the number of **Unique Learners**. The difference between these two metrics reveals how many learners are applying for multiple opportunities.
- **Enrollment Rate**: The relationship between the **status** column and the total number of applications is used to calculate the **Enrollment Rate**, a key measure of conversion success.
- **Age at Application**: The **age_at_apply** is derived from the **birthdate** of the learner and the **apply_date**. This is crucial for demographic analysis and for ensuring compliance for underage applicants.

By understanding these relationships, we can effectively create a dynamic and insightful dashboard that allows for filtering, grouping, and in-depth analysis of our data.

4. Mapping Table Walk-Through

A mapping table is a crucial data structure designed to provide a structured and organized understanding of any dataset. It systematically categorizes various properties of the data—such as Key Performance Indicators (KPIs), Metrics, and Dimensions—to create a well-defined framework. This framework is essential for building dashboards and visualizations that accurately represent both numerical and categorical information. The mapping table outlines the properties of a dataset using meaningful labels, logical groupings, and visual elements, ensuring accurate and

user-friendly visualizations. It helps to clarify table properties, provides a logical approach to data interpretation, and supports the derivation of meaningful insights.

Master Table – Mapping Table Overview

The mapping table serves as the foundational blueprint, defining how raw data from the master_table—which details relationships between learners and various opportunities—is transformed into meaningful, visual insights within a dashboard. Each field in the mapping table is meticulously categorized, potentially transformed, and assigned a specific business purpose. This section explores the roles and underlying logic of each group of fields.

Dimension Fields – Filtering, Grouping, and Descriptive Context

Dimension fields are fundamental for segmenting, categorizing, and filtering data across the dashboard. They enable users to explore data through relevant attributes such as learner demographics, geographic location, and opportunity classifications.

1. **Learner ID and Opportunity ID:** These fields act as unique identifiers, crucial for establishing clear relationships between learners and the opportunities they engage with. They also facilitate seamless integration with other datasets if necessary.
2. **Email:** While not used for direct data analysis, the email field is included for potential reference and communication purposes.
3. **Degree, Major, and Institution:** These provide vital academic and demographic context about learners, enabling comparisons based on educational backgrounds and institutional affiliations.
4. **Opportunity Name, Opportunity Code, and Category:** These fields enhance the filtering and lookup capabilities for various types of opportunities, including internships, courses, or events.
5. **Assigned Cohort:** This dimension is used to group learners into predefined batches, which is essential for tracking cohort performance and managing capacity.
6. **Gender, City, State/Region, and Country:** These provide demographic and geographic filters, allowing for the segmentation of learners based on identity and location.
7. **Status:** This field acts as a key filter for learner engagement, typically indicating whether a learner is currently active or inactive within the platform.

All dimension fields are primarily textual and generally do not require complex transformations. However, they are vital for enabling interactive filtering and drill-down capabilities in the dashboard.

Metric Fields – Measuring Time and Quantity

Metric fields represent numeric or time-based values that quantify learner activity or program characteristics. They form the basis for trend analysis, performance tracking, and temporal comparisons.

1. **Apply Date, Start Date, and End Date:** These are critical date fields that facilitate timeline analysis, allowing for the tracking of application cycles and the monitoring of opportunity durations.
2. **Duration:** Derived as the difference between the start and end dates, this metric indicates the length of a program or opportunity. It often serves as a direct KPI in time-based evaluations.



- 3. **Size:** Refers to the capacity of each opportunity—the number of learners it can accommodate. This helps in assessing resource allocation and planning.
- 4. **Birthdate:** This field is used not for direct analysis, but specifically to calculate the learner's age or age group at the time of application.
- 5. **User Create Date:** This metric helps monitor platform usage over time by indicating when learners first joined or were onboarded.

Although based on simple data types like dates or numbers, these metrics unlock powerful time-series visualizations, trend analysis, and duration-based KPIs within the dashboard.

KPI Fields – Aggregated Business Insights

Key Performance Indicators (KPIs) are derived from the combination and aggregation of dimension and metric fields. They represent high-level business metrics and performance insights that the dashboard is designed to convey.

- 1. **Total Enrollments/Applications:** Calculated by counting the number of learner-opportunity combinations, this KPI measures the overall application volume across the platform.
- 2. **Total Unique Students:** This counts distinct learner IDs, providing a snapshot of the platform's reach in terms of unique users.
- 3. **Most Popular Course:** Identifies the opportunity (course) that has received the highest number of applications, highlighting what resonates most with learners.
- 4. **Most Occupied Cohort:** Pinpoints the cohort with the maximum number of assigned learners, which can signal demand or potential capacity issues.
- 5. **Cohort Duration Metrics (maximum, minimum, and average):** These provide insight into the variability of program lengths, aiding in planning and standardization.
- 6. **Cohort Size Metrics:** Evaluate how large or small different opportunity groups are, offering visibility into resource allocation.
- 7. **Top City by Learners Applied:** Surfaces the city with the highest number of applicants, useful for targeted outreach or regional planning.
- 8. **Application Rate Over Time:** Calculated as a ratio of applications to learners over time, highlighting growth or decline in engagement across time intervals.
- 9. **Age Group Distribution:** Derived by calculating learners' ages at the time of application and then grouping them, providing insights into which age demographics are most engaged.
- 10. **Average Monthly Applications:** Shows how applications trend month-by-month, giving program managers insight into seasonality or the impact of campaigns.

Each KPI is derived through a combination of grouping, filtering, and aggregation techniques (such as COUNT, AVG, MIN, MAX, or DISTINCT), using fields like apply_date, birthdate, opportunity_id, and learner_id.

Transformation and Business Logic

Most fields in the mapping table are utilized in their raw form, requiring minimal transformations. The primary transformations include:



- 1. **Calculations of Age and Age Groups:** Determined by finding the difference between `apply_date` and `birthdate`.
- 2. **Duration:** Calculated as the number of days between `start_date` and `end_date`.
- 3. **Aggregated KPIs:** Involve standard SQL-style operations such as `GROUP BY`, `COUNT`, or `AVG`.

These transformations are typically implemented within the backend logic of the dashboard or at the data preparation layer.

Purpose and Value of the Mapping Table

This mapping table ensures a structured and consistent approach to dashboard development and data interpretation. It serves several key purposes:

- **Clear Linkage:** Establishes a transparent connection between raw data and actionable business insights.
- **Intuitive Exploration:** Enables intuitive visual exploration through clearly defined dimensions and filters.
- **Performance Indicators:** Provides essential performance indicators for informed decision-making across learner engagement, program popularity, and operational metrics.
- **Scalability:** Supports scalability by simplifying the addition of new metrics or views without introducing ambiguity in data interpretation.

5. Wireframe Planning & Designing

Importance of Wireframe:

A wireframe is a fundamental visual guide used in design to illustrate the layout and structure of a digital interface, such as a website, application, or dashboard. It delineates the placement of various interface elements—including buttons, charts, menus, and text—without incorporating detailed design aspects like colors, typography, or images.

Wireframes are typically developed during the initial stages of a project to plan functionality and user experience (UX) prior to advancing to detailed design and development. They can be created manually or through digital design tools such as Figma, Balsamiq, or Adobe XD. In the context of data visualization dashboards, wireframes provide stakeholders with a clear understanding of:

- The key metrics and visualizations to be displayed.
- The placement of filters, charts, KPIs, and tables.
- The ways in which users will interact with the data.

Wireframes are essential for ensuring team alignment regarding layout, navigation, and content hierarchy before the complete interface is implemented.

Identifying Key Dashboard Section:

1. KPI Boxes (Top Row of Dashboard)

These are quick-glance scorecards placed at the top of the dashboard, allowing stakeholders to immediately see the most important metrics without digging into charts or tables.

Key KPI Scorecards:

KPI	Example Value	Business Use	Visual Representation
Total Applications	113,602	Shows overall demand across all opportunities	Single large number indicator
Unique Learners	57,966	Captures true reach, avoids double	Distinct learner count card

KPI	Example Value	Business Use	Visual Representation
		counting	
Applications per Opportunity	Varies by program	Highlights program popularity & top-of-funnel demand	Average or dynamic number card
Enrollment Count	76,109	Represents successful conversions into enrollments	Enrollment number card
Enrollment Rate	67%	Tracks funnel effectiveness (applications → enrolled)	Percentage scorecard

These 5 KPI boxes should be placed in the top row of the dashboard, each as a clean number card/scorecard with minimal text, so users can grasp the overall performance at a glance.

2. Graph Placements (Main Dashboard Visuals)

Placed directly **below the KPI boxes**, these visuals provide detailed insights into application trends, distributions, and breakdowns. A **mix of chart types** ensures both clarity and engagement for different types of analysis.

Visual Type	Chart Title	Purpose / Insight	Placement Suggestion
Line Chart	Enrollment trend Over Time	Tracks on how enrollment change over different time periods, highlighting peaks/troughs.	Top-Center (below KPIs)
Pie Chart	Distribution of Applications by Gender	Displays proportion of applications by gender, providing demographic insights.	Bottom-Center
Stacked Bar Chart	Applications per opportunity	Compares applications across different opportunity_id, highlighting dominant applicant groups.	Top-left
Area Chart	Application by Age	Compares applications across different Age, highlighting dominant applicant groups.	Top-right(below KPI's)
Row Chart	Top 5 countries of application	Compares top 5 countries across different number of application	Bottom-left
Donut Chart	Status of countries, gender, age	Compare status of countries, age and gender	Bottom-right

With this setup:

- a) Top row → KPIs (scorecards)
- b) Second row → Stacked bar, Line chart & Area chart
- c) Third row → Row chart, Pie chart & Donut chart

This creates a clean and logical funnel of insights:
Quick metrics → Trends & distribution → Demographics & group analysis.

The visualizations (using “Cleaned_Master_Table.csv”) which are going to be inside the Wireframe are below :

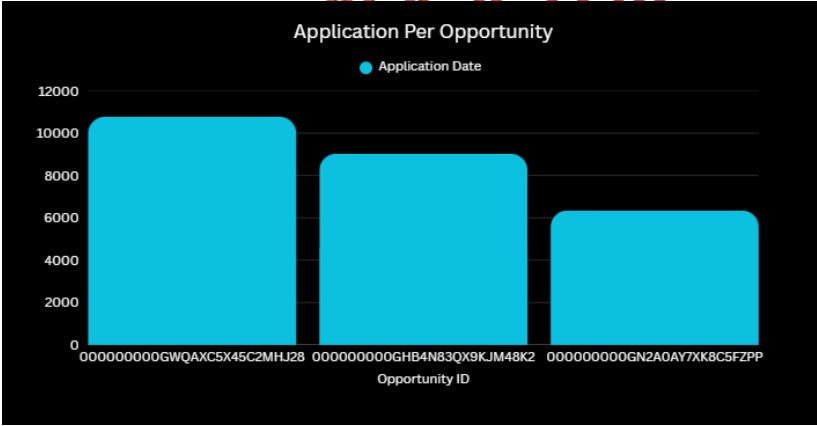


Figure-1: Stacked bar Chart Of Application per opportunity

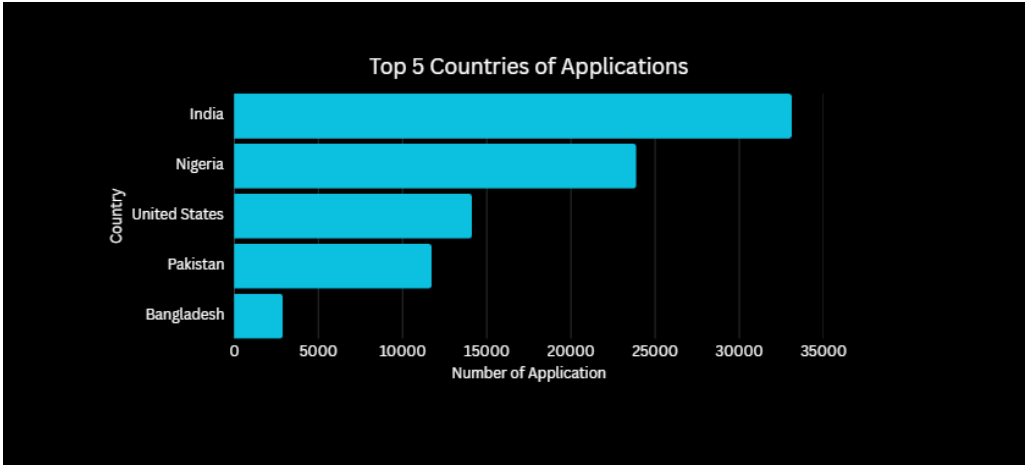


Figure-2: Row Chart of Top 5 Countries by number of application

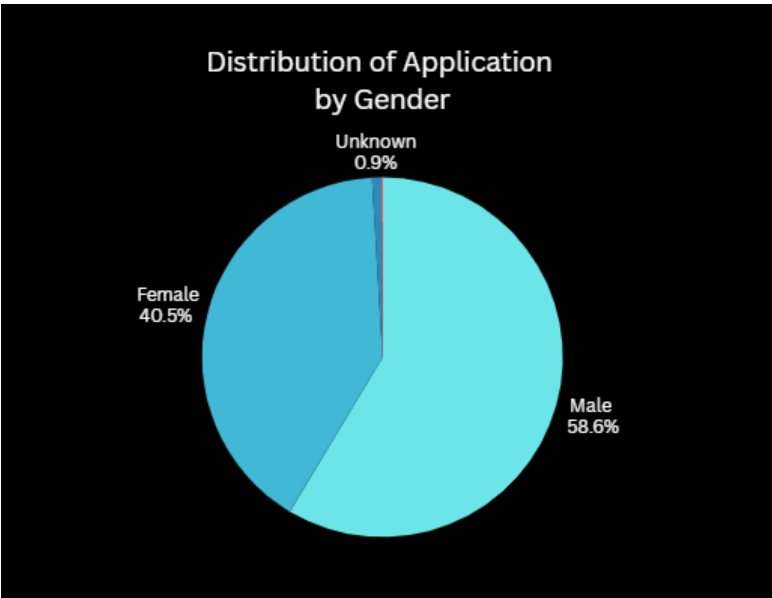


Figure-3: Pie Chart Distribution Of Applications by Gender

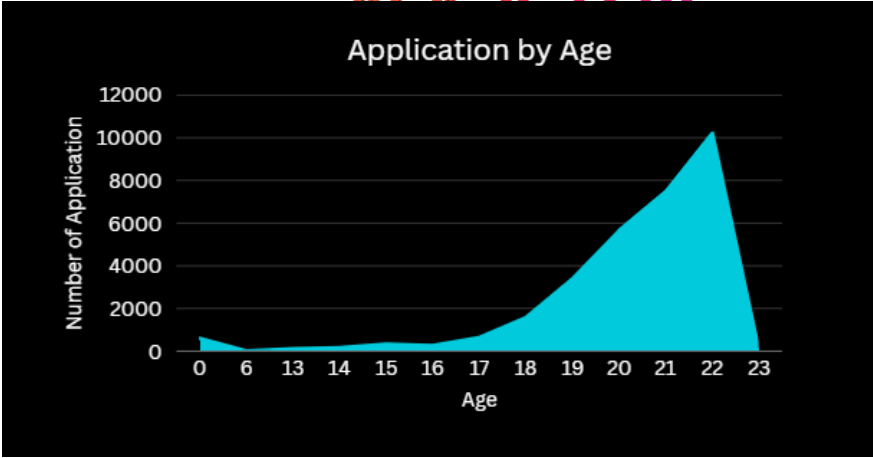


Figure-4: Area Chart Of Applications by Age Group

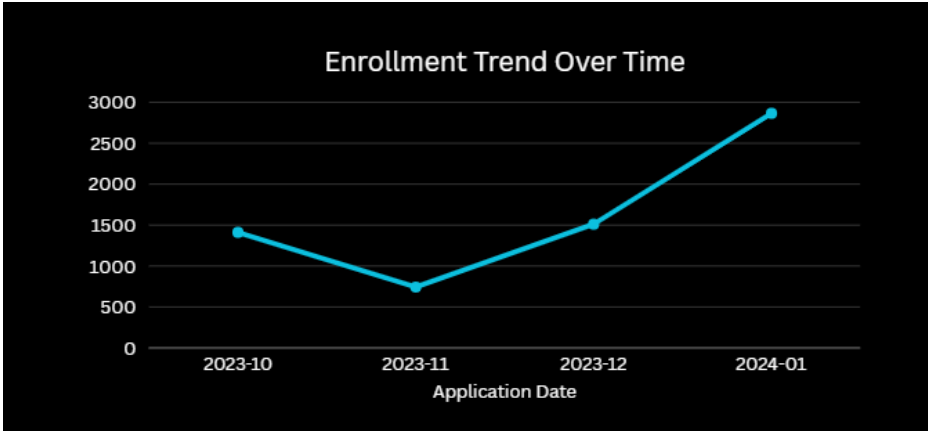


Figure-5:Line Chart of Enrollment Trend over Time

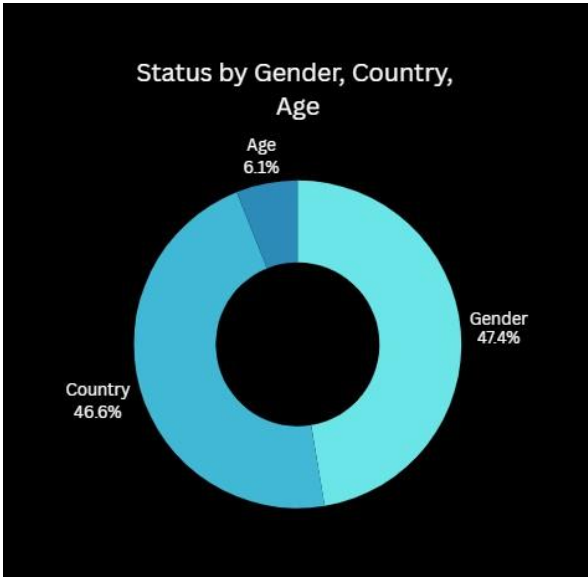


Figure-6: Donut Chart of Status by gender, age & country

3. Filters and Drill-Down Controls

To make the dashboard **interactive and user-driven**, filters and drill-downs should be positioned at the **top-right or left-hand side** of the interface. Based on the dataset fields, the following controls are recommended:

Filter Name	Source Field (from dataset)	Purpose
Course / Opportunity Name	opportunity_id, opportunity_name	Narrow down KPIs and visuals by specific courses/opportunities.
Department / Category	category	Analyze applications and enrollments by academic department or program type.
Batch / Cohort	cohort_start_date, cohort_end_date	Filter learners by intake period for batch-wise comparisons.
Gender	gender	See demographic splits across KPIs and charts.
Time Period	apply_date, user_create_date	Select a date range to view applications/enrollments over time.

This setup ensures Filters give users control over the dataset slice they want

6. Final Layouts & Annotations of our Wireframe



7. Wireframe Annotations

Section	What Does This Represent?	Why This Chart / Placement?
KPI Boxes (Top Row)	Quick-glance metrics: Total Applications, Unique Learners, Enrollments, Enrollment Rate	Why is this KPI here? Placed at the top for immediate visibility, giving stakeholders an instant performance snapshot.
Enrollment trend Over Time (Line Chart)	Shows the trend of Enrollment across different time periods	Why this chart type? A line chart is ideal for spotting peaks, drops, and seasonal patterns over time.
Top 5 countries of application(Roe Chart)	Displays regional distribution of applications	Why this chart type? Row (horizontal) charts are used because they make long category labels (like countries or institutions) easier to read and compare.
Applications by Gender (Pie Chart)	Shows the proportion of applications by gender	Why this chart type? A pie chart intuitively communicates demographic splits in simple proportions.
Applications per opportunity (Bar Chart)	Compares applications across different opportunity groups	Why this chart type? A bar chart is best for categorical comparisons, highlighting dominant age groups.
Application by age (Area Chart)	Shows applications across different Age, highlighting dominant applicant groups.	Why this chart type? Area charts are used to show trends over time while also emphasizing the magnitude (volume) of values.
Status by Countries ,Age & Gender(Donut Chart)	Shows status of countries, age and gender	Why this chart type? Donut charts are used to show the proportion of categories as parts of a whole while keeping the center open for highlighting totals or key values.
Filters & Drill-Down Controls (Side Panel)	Dropdowns for course, department, batch, gender, and time period; drill-down for deeper details	What will filters help users do? Enable interactivity by slicing data; drill-downs allow users to click and explore more details at the category level.

8. Conclusion

Aspect	Key Points
Foundation of the Dashboard	- Mapping Table Walk-Through provides structured understanding of the dataset.- Defines dimensions, metrics, and KPIs.- Transforms raw master table data into actionable insights.
Data Organization	- Categorizes identifiers, demographic fields, program attributes, and aggregated performance measures.- Creates logical framework for accurate visualizations.- Simplifies data interpretation and supports drill-down analysis.
Dashboard Design (using “Figma”)	- KPI boxes highlight key metrics (total applications, unique learners, enrollment rates).- Line charts, maps, pie charts, and bar charts show trends, geography, and demographics.- Filters and drill-down controls allow segmentation by course, department, cohort, gender, or time period.
User Experience & Interactivity	- Dashboard flows logically from summary metrics to detailed breakdowns.- Supports exploration of specific data points for informed decision-making.
Accuracy & Consistency	- Mapping table ensures data accuracy and consistency.- Wireframe layout prioritizes usability and insight discovery.
Overall Impact	- Provides robust framework for monitoring learner engagement, program demand, and operational effectiveness.